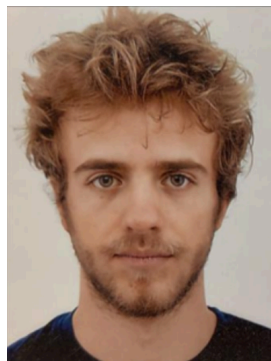


Capstone Project

A qualitative analysis of Hanoi Air quality



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Abstract

Air pollution has become a major concern for people living in large cities, especially families with young children and the elderly, who are particularly sensitive to air quality. Hanoi, due to its rapid development, growing population and industrialization, faces some of the highest pollution levels in Vietnam. In this project, we will analyze the factors influencing air quality in Hanoi over the years.

Key words: Air quality analysis, Atmospheric data, OpenMeteo API, PM2.5, PM10

Project motivation

While we (international students) were strolling in the city of Hanoi, we soon noticed the ambient smog and strong smell of fuel. Most of us came from Europe and were surprised to see such a stark change in air quality, as this was something we took for granted back home. Therefore, in order to have better insights into the health risks and severity of Hanoi's air pollution, we decided to choose this topic as our Project Capstone. To establish the relevance of this project, let us first consider Hanoi's air quality in comparison to other cities worldwide.

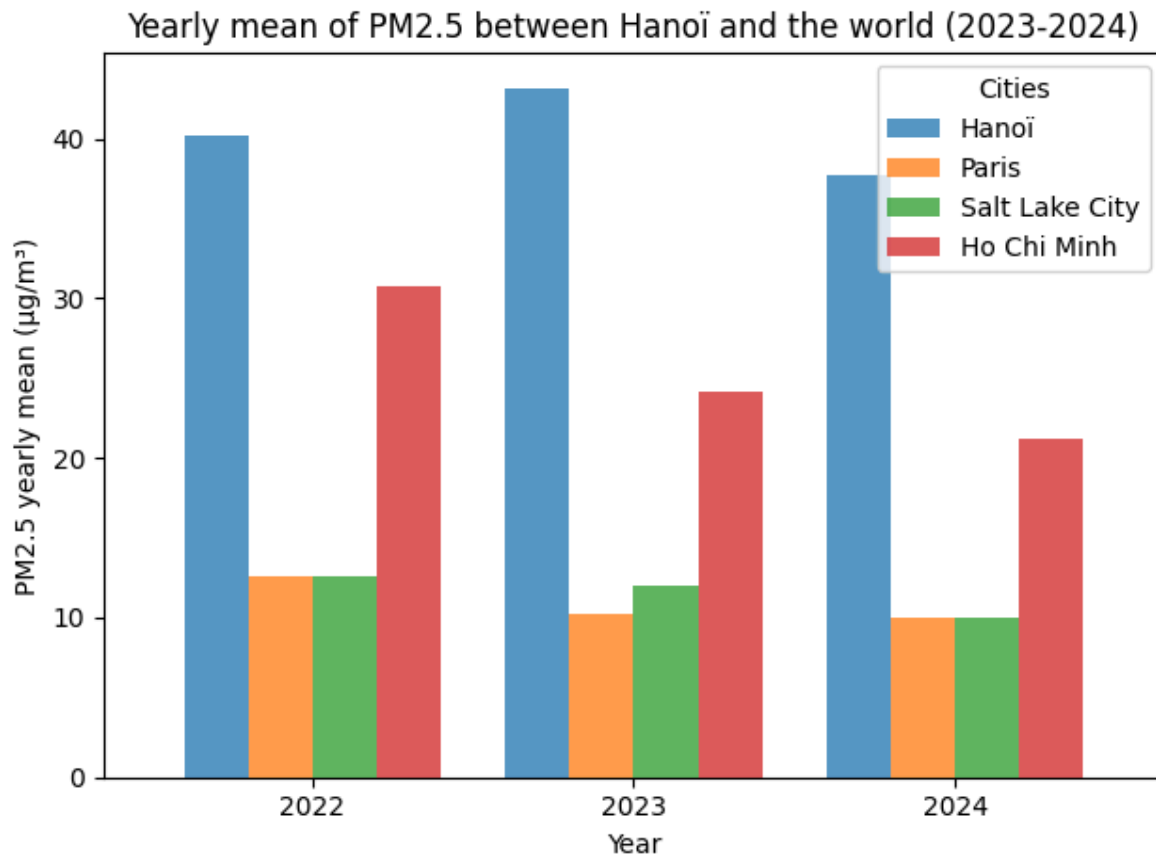


Figure 1 : Yearly mean PM2.5 comparison between Hanoi and the world

Figure 1 shows the yearly PM2.5 concentrations in several cities around the world, comparing air quality in Hanoi with other major urban centers.

It is alarming to see cities like Salt Lake City and Paris maintain much lower PM2.5 levels throughout the year compared to Hanoi. This comparison highlights the stark differences in air quality worldwide, with Southeast Asian cities, such as Hanoi and Ho Chi Minh City, displaying higher average pollution than cities in North America and Europe.

This comparison underscores the need for ongoing efforts to manage pollution in rapidly urbanizing areas like Hanoi, while the data from lower-pollution cities could offer insights into successful pollution control strategies that may be adaptable to Hanoi's context.

1. Introduction

Rapid population growth and industrialization are worsening air quality in major cities globally, highlighting the need to analyze pollution and its seasonal and geographical variations. Such analysis enables authorities, scientists, and urban planners to identify key factors and times of year when air pollution peaks, informing strategies to mitigate its impact. This project focuses on Hanoi, the capital of Vietnam, where exponential growth in population, industrial activities, and vehicular traffic contributes to significant air quality issues. We chose to examine patterns and trends in particulate matter PM2.5 because this matter seems to be the most widely studied but also the most dicey. It is indeed a major pollutant linked to proven respiratory and cardiovascular health risks. This project seeks to better understand the climatic, geographical, and human influences driving pollution in Hanoi.

Using historical air quality data collected between November 2022 and November 2024, we therefore try to explain the rise and fall in PM2.5 levels from various causes, such as seasonal, geographic, or temporal factors. The data was retrieved through public APIs and analyzed using various data processing and visualization techniques. Key objectives include identifying peak pollution periods, comparing Hanoi's air quality with that of other major cities, and examining the effects of meteorological variables, such as temperature and humidity, on PM2.5 concentration.

This project is therefore particularly useful for informing residents and authorities regarding outdoor activity schedules, awareness periods, and pollution reduction efforts, in order to create healthier habits for the inhabitants of the Vietnamese capital. It also provides an insight to sanitaries authorities to know where it is paramount to operate in order to be the most effective, in an effort to reduce air pollution.

2. Hanoi Air Quality Dataset

2.1 Data collection

At first, we struggled to find free source sites with enough data in order to effectively highlight global trends through the years. After a few hours of browsing, we finally found a couple of reliable sources of data which were : Open Meteo, IQAir, and OpenAQ.

For detailed analysis, we collected air quality data from these multiple sources (Open Meteo, IQAir, and OpenAQ) using their respective APIs. These sites provided hourly and daily pollution level data, allowing us to monitor long-term changes as well as precise change through the day. Although we faced limitations in the free availability of historical data, especially for records prior to 2022, we kept our analysis period from January 2022 to the present. This relatively small period still allowed us to examine recent trends and variation in PM2.5 levels.

These sites provide reliable data, from PM2.5 and PM10 probes all around the world, from Washington to Paris, to Hanoi. All that is necessary is to have the coordinates (latitude, longitude) of the city you want to measure the air pollution in.

2.2 Data Preprocessing

To begin with, we gathered our data by using a HTTP request through an API. After doing so with our 3 APIs, we had to process them before using them for analysis. However, we soon realized that the 3 APIs gave us 3 times the same data. Thus, we decided to only keep the OpenMeteo database and API for this study. Indeed, it had a better precision and a larger data range than the other databases. We sorted out the data from this API by each pollutant : in one hand PM2.5 and in the other, PM10. We also sorted the data out by hour, using the panda library. Finally, we used Python, and in particular, we used our matplotlib library knowledge to draw graphs.

2.3 Exploratory Data Analysis

In this section, we will conduct an exploratory data analysis to gain a comprehensive understanding of the dataset, examining patterns and key factors that influence air quality in Hanoi. This analysis aims to provide valuable insights into the main causes of pollution in the city.

2.3.1 Factors affecting the air pollution trend in Hanoi

Before getting started, it's essential to address the constant sources of air pollution that contribute to steady PM2.5 emissions throughout the year. One of the primary sources of pollutants is industrial activities. Indeed, industries in and around Hanoi are critical sources of pollution, especially in areas with factories and manufacturing plants. Because the city of Hanoi is highly populated, traffic congestion makes vehicular emissions one of the main contributors to urban pollution : it releases not only PM2.5 and PM10 but also nitrogen oxides, which further contribute to air deterioration through chemical reactions in the atmosphere. To cap it all, a notable cultural habit contributing to pollution, albeit on a smaller scale, is the practice of burning waste materials in residential areas. Emissions from these 3 sources contribute to both PM2.5 and PM10 levels year-round. In figure 1, you can see the global PM2.5 concentration through the year in Hanoi, which is, on average, extremely high. But what is even more striking are the numerous spikes and low of this concentration, throughout the year.

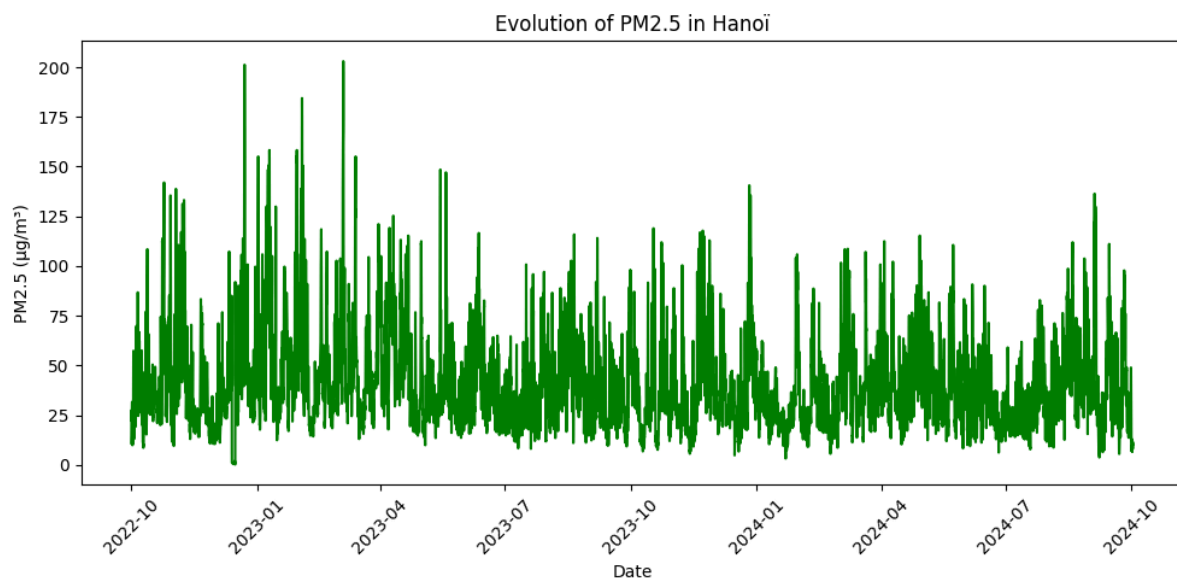


Figure 2 : Last 2 years PM2.5 air concentration in Hanoi

Now that this baseline is set, we can examine the causes of highs & lows throughout the year.

Seasonal Changes

First of all, let's tackle the impact of seasonal changes.

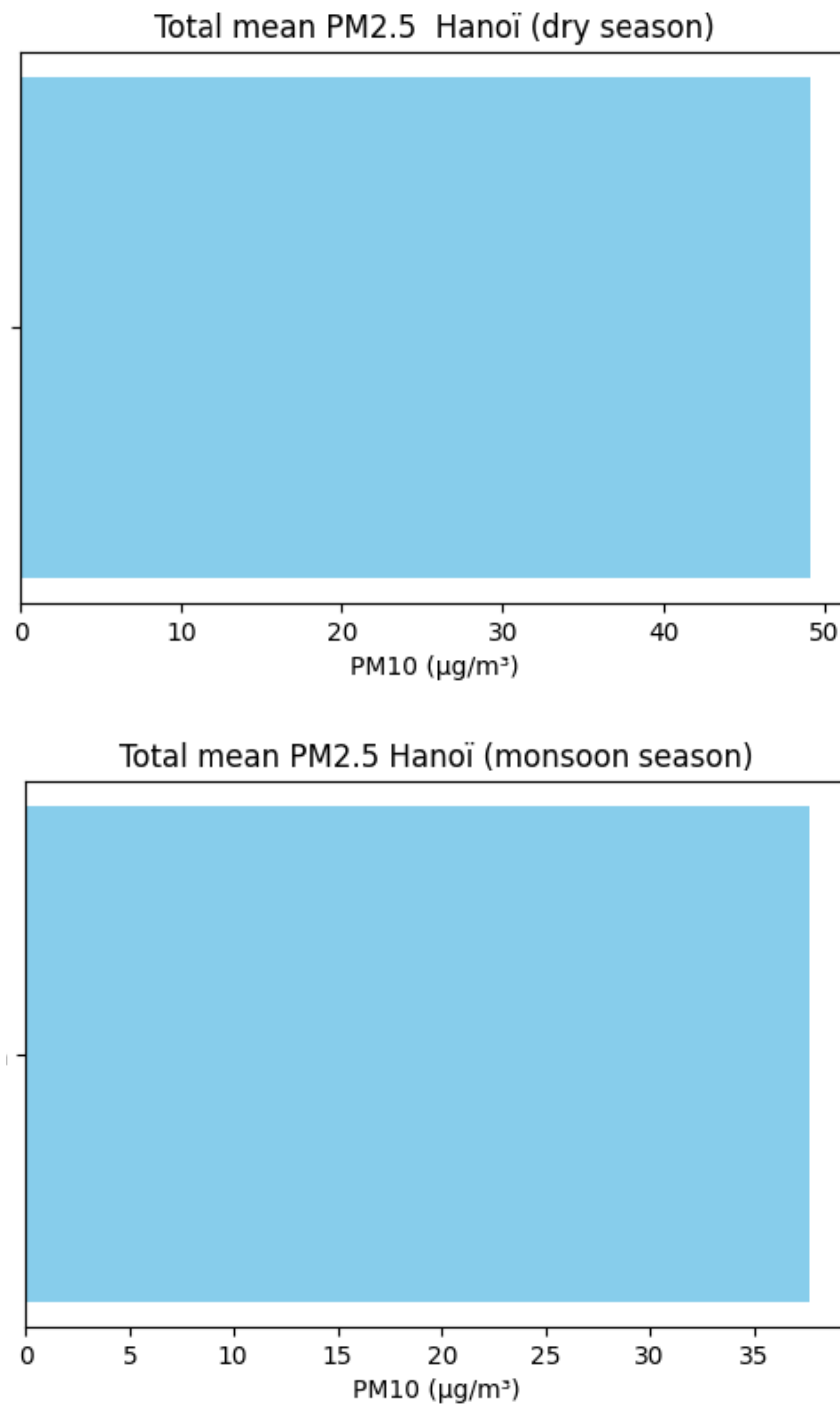


Figure 3 : Total mean PM2.5 in Hanoï during dry and monsoon season

Firstly, we can see on Figure 3 , which displays the average PM10 concentration through the monsoon or dry season that PM10 concentration is higher during the dry season compared to the monsoon season. Let's try to understand the various phenomena that play a role in PM2.5 and PM10 concentration and detail the most prominent ones.

Rain & Wind

During the monsoon season, rainfall plays a natural cleansing role, reducing pollutant concentrations as rain washes particulate matter out of the atmosphere. However, during colder months (of the monsoon season), lower temperatures also come with reduced wind speeds, allowing pollutants to accumulate in the atmosphere.

Due to the lack of rainfall during the dry season and the greater stability of the atmosphere, pollutant concentrations generally increase, allowing pollutants to stay closer to the ground.

Nonetheless, the lack of air circulation can cause air stagnation, thus making it difficult for pollutants to spread out, thereby keeping high PM2.5 and PM10 concentration in the city, especially in certain areas. Without further ado, let's try to **illustrate** the impact of wind and rain on the air quality of Hanoi.

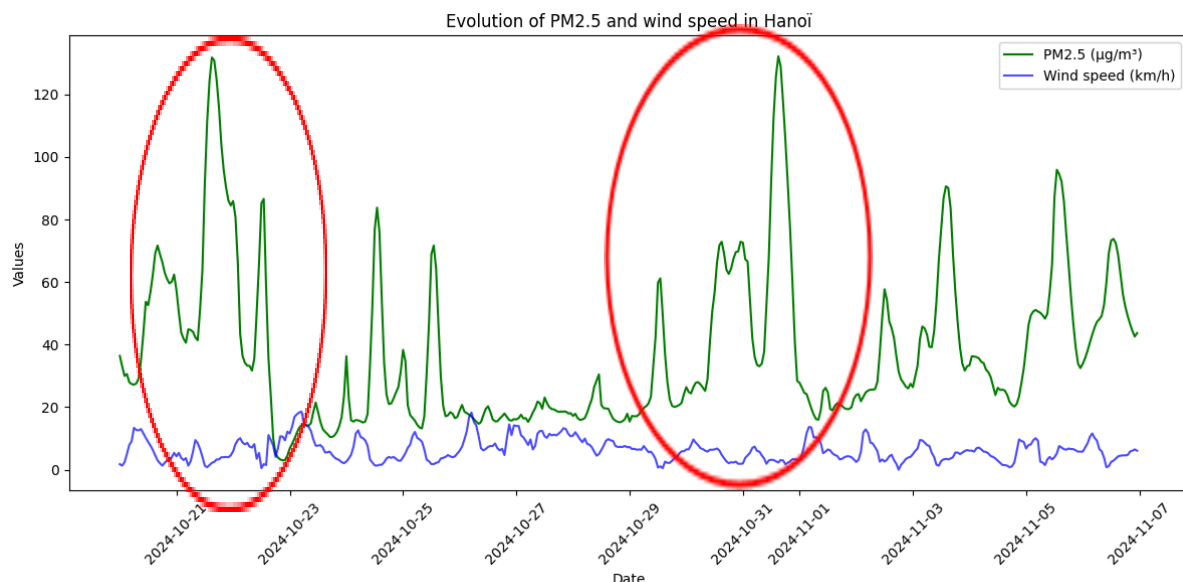


Figure 4 : Relation between PM2.5 concentration and the wind speed in Hanoi

As observed, each time wind speed increases significantly, pollutant concentrations decline right after. For instance, on October 23, concentration levels dropped from over $120 \mu\text{g}/\text{m}^3$ to below $20 \mu\text{g}/\text{m}^3$. A similar pattern occurred on November 1, with concentrations falling from more than $120 \mu\text{g}/\text{m}^3$ to around $20 \mu\text{g}/\text{m}^3$.

Moreover, on some occasions, multiple consecutive days of high wind speed, like the timeframe from October 26 to October 29 with an average speed of 12 km/h , correlate with persistently low pollutant levels in Hanoi.

Conversely, when wind speed is at its lowest, pollutant concentration tends to peak. On October 21, for example, the concentration exceeded $120 \mu\text{g}/\text{m}^3$ as wind speed dropped to nearly 0 km/h .

However we can see on some occasions that wind speed doesn't have a significant impact on PM2.5 concentration. For instance, when looking at the period spanning from November 3rd to 7th, wind speed doesn't seem to correlate in any way with the pollution peak. We can reasonably think that wind has an impact on the air quality of Hanoi. Let's see if our hypothesis is right or not.

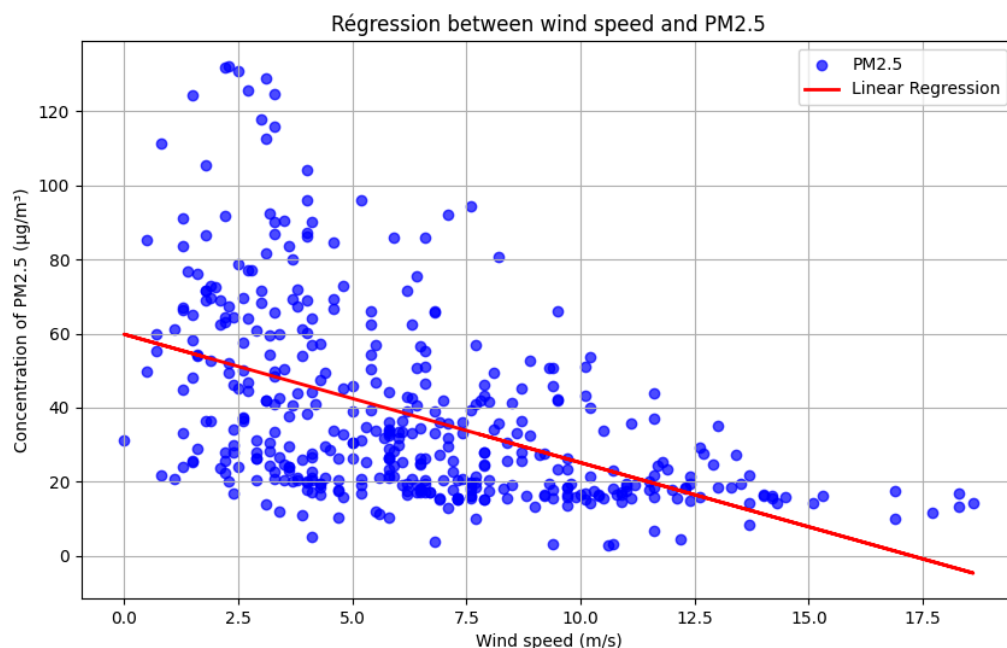


Figure 5 : Regression between wind speed and PM2.5

Linear regression is a fundamental statistical method used to explore and quantify the relationship between two variables, providing insights into how one variable influences the other. By performing a linear regression analysis between wind speed and PM2.5 concentration, we can quantify the strength and direction of their relationship, in order to understand better how changes in wind conditions affect air quality.

After drawing the linear regression, we can see on *Figure 5* that our hypothesis about a strong bond between the wind speed and the concentration of PM2.5 seems right. The slope of the linear regression is -3,4, meaning that, on average, for every m/s of wind, the PM2.5 concentration decreases by $3,4\mu\text{g}/\text{m}^3$. This linear regression therefore confirms our previous hypotheses and clearly shows a decrease in PM2.5 concentration as wind speed increases. This validates the idea that stronger winds promote the dispersion of fine particles, thereby reducing their accumulation in the air in Hanoi.

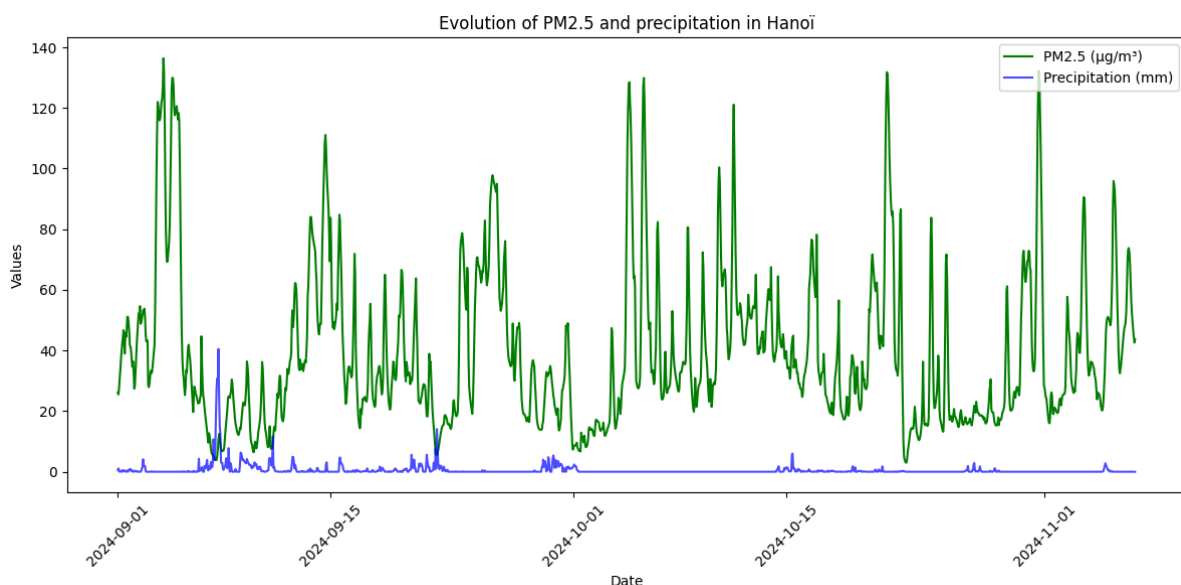


Figure 6 : Relation between the precipitation and the PM 2.5 concentration in Hanoi from September to November

Concerning rainfall and its PM2.5 cleansing role, we can study *Figure 6*, covering the period from September to November, in rainfall and PM2.5 concentration. We can see the rain precipitation roughly corresponds to drop in PM2.5 concentration, indicating a correlation between the two.

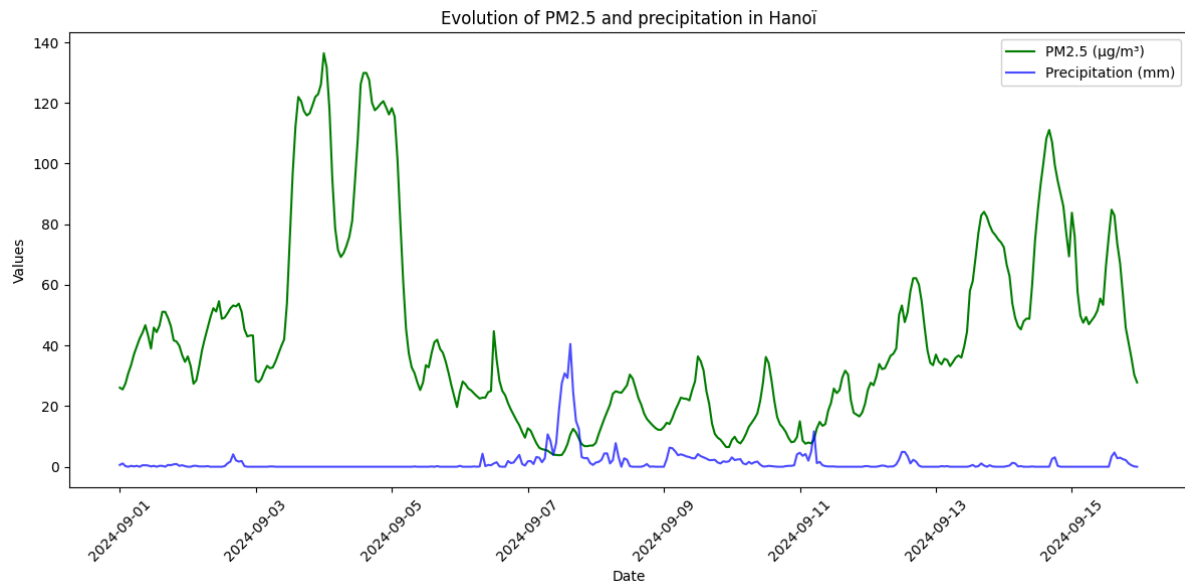


Figure 7 : Evolution of PM2.5 and precipitation in September

Indeed, we can see on *Figure 7* that when a prominent surge in rain precipitation happens around the 7th of September, concentration in PM2.5 substantially decreases to a point where it is approximately healthy (less than $12 \mu\text{g}/\text{m}^3$ for a 24hrs period). This illustrates the power of rain in reducing the PM2.5 concentration in Hanoi.

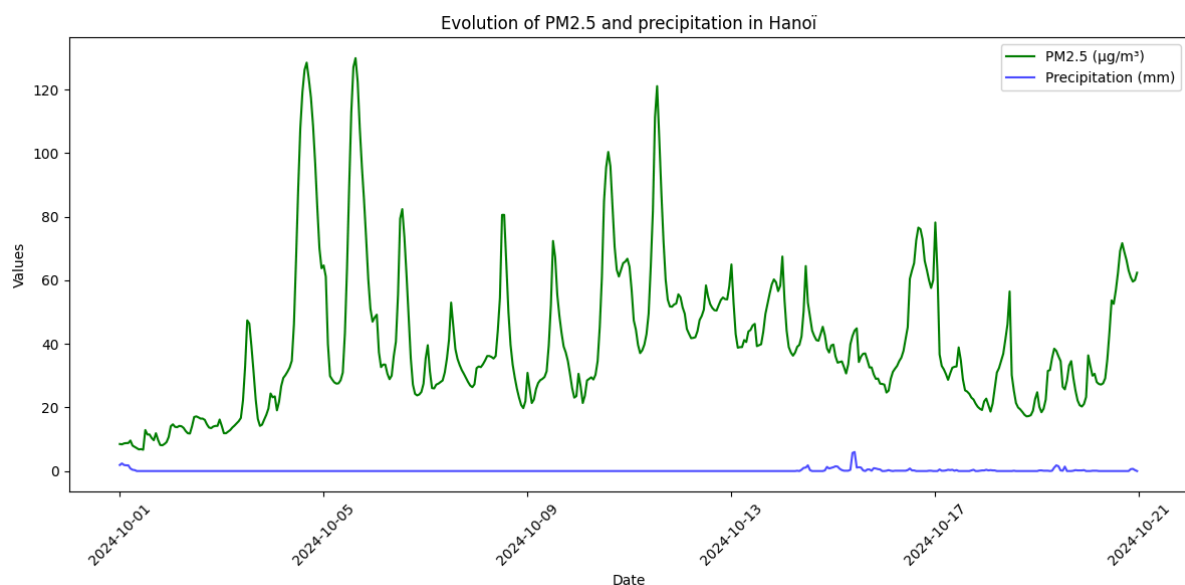


Figure 8 : Relation between the precipitation and the PM 2.5 concentration in Hanoi during October

Conversely, we can also highlight what happens in case of lack of rain. To illustrate this, we can look closely at the beginning of October 2024 on *Figure 8*. We can observe that when precipitation is scarce, PM2.5 air concentration doesn't have a ceiling and can go as high as $120\mu\text{g}/\text{m}^3$ in the daytime peak. This again strongly evidence the correlation on a given period between air pollution ceiling and rain patterns.

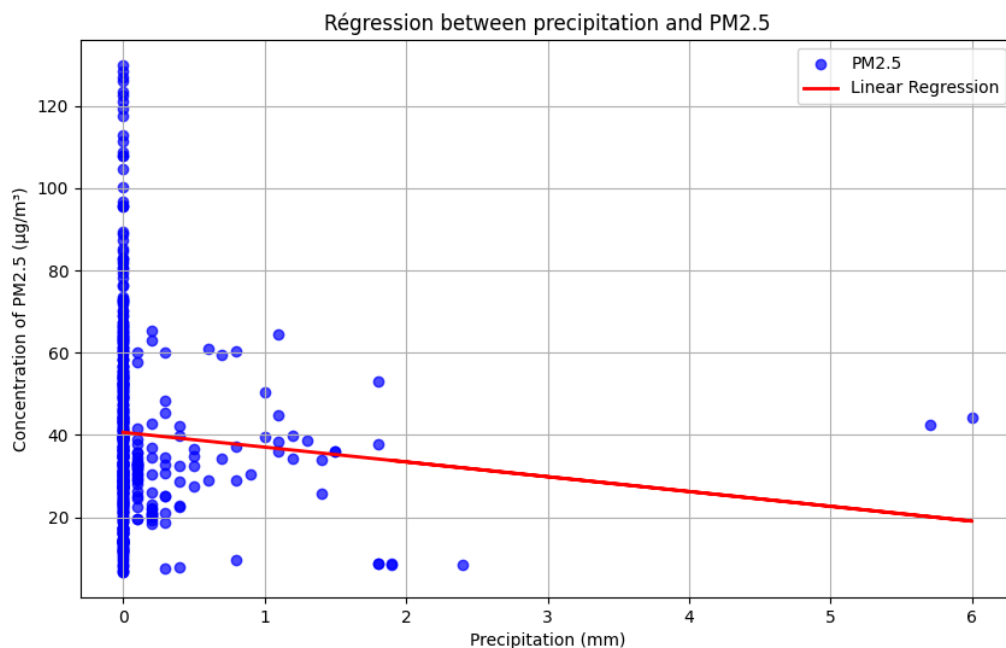


Figure 9 : Regression between precipitation and PM2.5

To assess our previous observations and their validity to a certain extent, we can evaluate the strength of the correlation between the two, using a linear regression. The figure represents a regression between precipitation and air pollution. We can see that there is a clear negative correlation between rain and air pollution, meaning that, on average, the more it rains, the less polluted the air is.

The slope is $-3,33$, meaning that, on average, for every mm of rainfall at a given time, the PM2.5 concentration is reduced by $3,33\mu\text{g}/\text{m}^3$. However, the rain precipitation rarely exceeds more than 1,5mm, and **the rain average is about 0,05mm** at any given time. Also, it is often equal to zero. In comparison, the wind is always here and with a **wind speed about 7,5 m/s on average**. This stark difference in average between the wind and the rain, even though both regression have the same slope, highlights the fact that wind play a much bigger role in reducing the PM2.5 concentration year round.

The impact of the wind is therefore extremely important in order to reduce the air pollution in Hanoi. In comparison, while the rain is similarly impactful on air pollution, its scarcity makes it that it has a very casual and situational effect on the air pollution.

Thermal Inversion

In the second place, let's tackle the role of thermal inversion phenomenon. Typically, air temperatures decrease with height above the Earth's surface, but inversions occur when the air near the ground cools more quickly than the air above it. This results in warmer air above cooler air, which acts like a cap, preventing much vertical movement in the atmosphere. Inversions usually set up in the early evening and last until a few hours after sunrise.

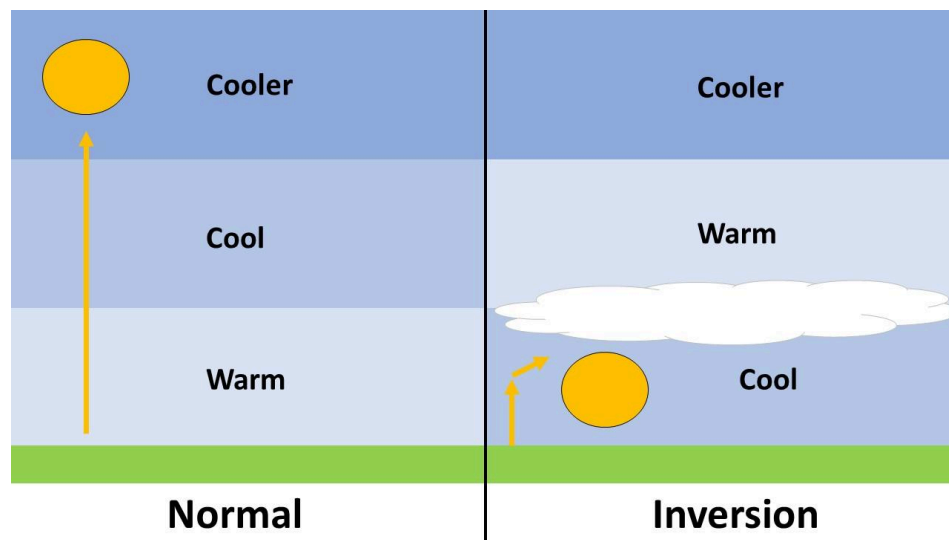


Figure 10 : Simplified drawing of thermal inversion

This phenomenon is especially common in the dry season, because there is less displacement of air in the atmosphere compared to the monsoon season where it rains, contributing to melding the different layers of air. Thermal inversion can lead to rapid increases in pollutant concentrations, as emissions from vehicles, industry, and other sources are confined near the surface.

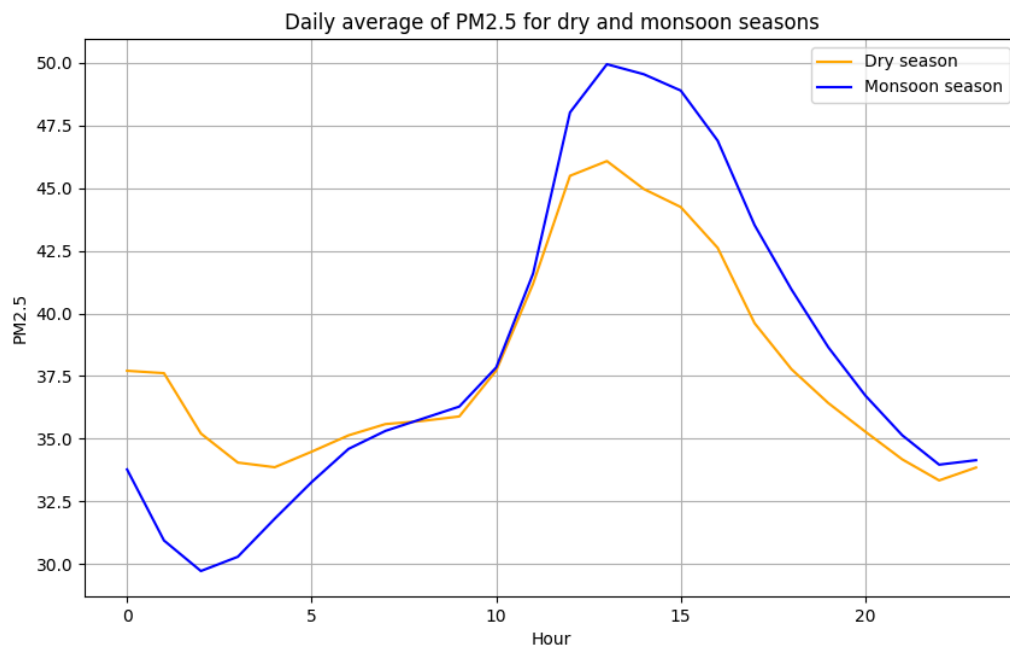


Figure 11 : Daily evolution of PMs' concentration

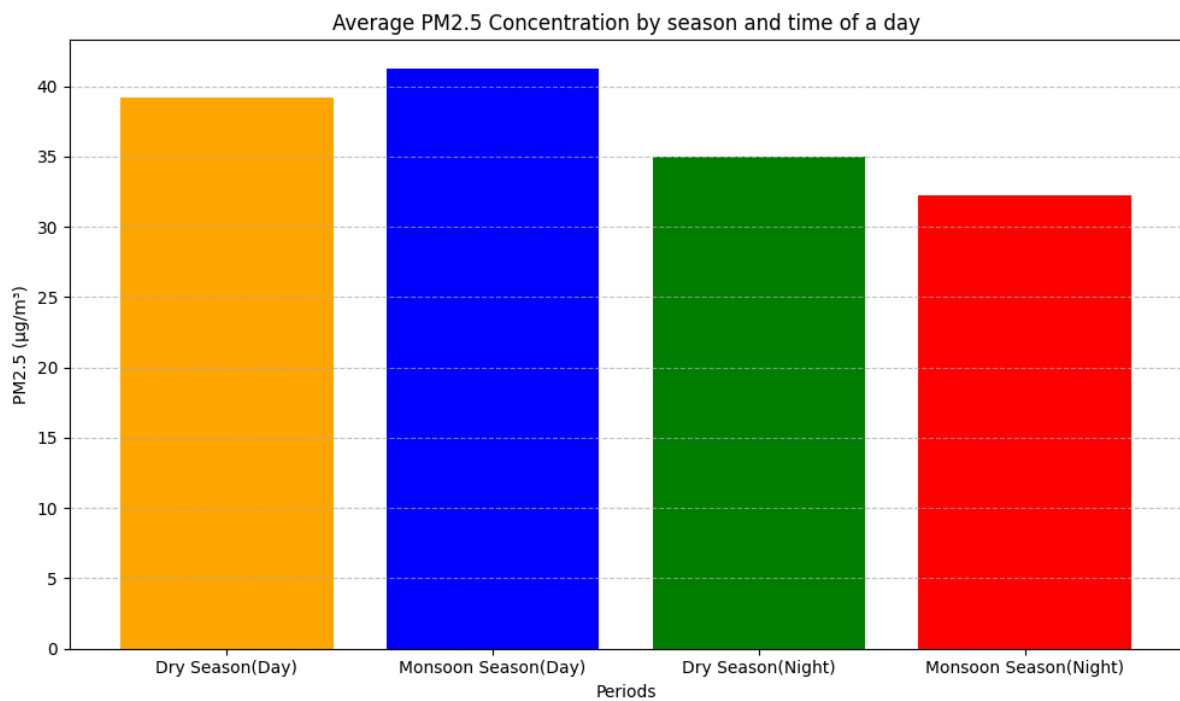


Figure 12 : Seasonal evolution of PMs' concentration

Let's try to highlight the thermal inversion phenomenon which mostly occurs during the dry season, during the night. In order to do so, we should compare the average PM2.5 concentration and the average nighttime reduction relative to daytime levels between a typical night in the dry season and a typical night in the monsoon season. Let's call 10pm - 6am a night, and 6am - 10pm a day.

For the figure 11, we computed the PM2.5 concentration during all dry season/monsoon season days or nights and put the average of each in a graph. Therefore each graph represents the average PM2.5 concentration depending on different times of the year, but also different parts of a day. The average concentration of the monsoon season nights is $32\mu\text{g}/\text{m}^3$, whereas the average concentration of the dry season nights is $35\mu\text{g}/\text{m}^3$. The average air pollution during the monsoon night is lower than that of the dry season. This result shows a slight evidence that thermal inversion is contributing to maintaining a high PM2.5 concentration through the night.

However, one might argue that this former method does not take into account the base PM2.5 during these two seasons. Therefore, comparing the monsoon and dry season nights and their PM2.5 average is not grounds to highlight an impact of thermal inversion on the air pollution. And he would be right : in fact, a higher PM2.5 concentration during the dry season nights may simply highlight a higher baseline concentration during the dry season globally.

For that reason, and in order to illustrate the impact of thermal inversion, it is relevant to examine the relative reduction compared to the average daytime PM2.5 concentration. Concerning the dry season days, as shown on Figure 12 the average daytime concentration is about $39,3\mu\text{g}/\text{m}^3$. With $35\mu\text{g}/\text{m}^3$, the night concentration is therefore 89% of what it is during the day, showing a 11% reduction. As for the monsoon season we can see an average daytime concentration of $41\mu\text{g}/\text{m}^3$. With $32\mu\text{g}/\text{m}^3$, the night air pollution is consequently 78% that of the daytime, showing a 22% decline.

This analysis therefore highlights the impact of thermal inversion in maintaining a higher air pollution during the dry season compared to the monsoon season. In fact, thermal inversion plays a role during the dry season in maintaining a high relative PM2.5 concentration through the night. Indeed, the relative air pollution reduction during the dry season night is 11% of what it is during the day, compared to 22% for the monsoon season.

We can therefore conclude that thermal inversion plays a prominent role in maintaining a high PM2.5 level during the night. It is however noteworthy to know that other minor factors may also play a role in the nightly air pollution during the dry season.

Agricultural burning

In the third place, we are going to discuss the impact of agricultural burning. Agricultural burning particularly during the post-harvest periods in nearby rural areas, contributes to Hanoi air pollution. Farmers often burn rice straw and other crop residues to prepare fields for the next planting season, especially from October to November, and in early spring. This practice leads to large amounts of particles and gases being carried to Hanoi by the wind.

Even if it's hard to precisely highlight the pollution induced by agricultural burning in Hanoi, the effects of agricultural burning are not limited to Hanoi but are part of a larger, regional pollution issue. This seasonal practice affects visibility and health across all northern Vietnam.

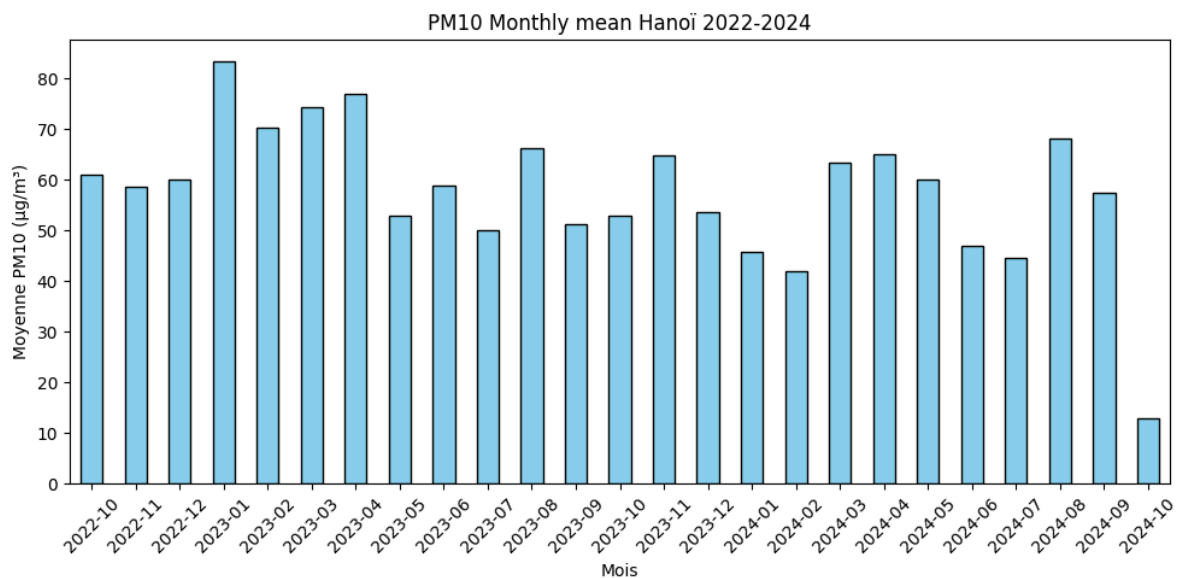


Figure 13 : Average concentration of PM10 per month during 2 years in Hanoi

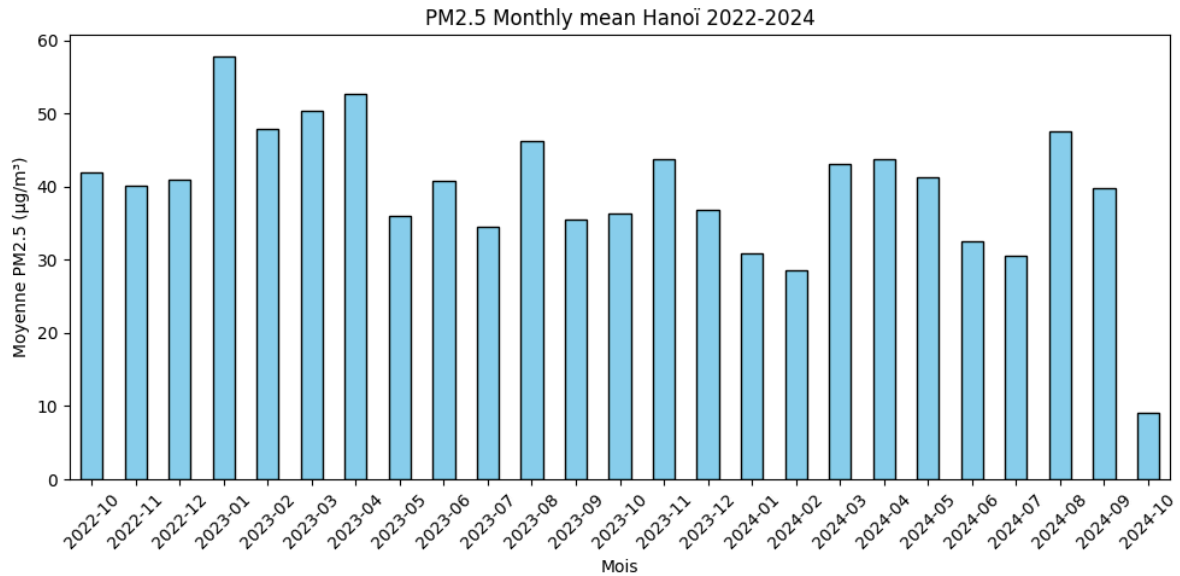


Figure 14 : Average concentration of PM2.5 per month in Hanoi

Trying to find a correlation between the rice straw burning and an hypothetical increase of PM2.5 (Fig 13) and PM10 (Fig 14) concentration in Hanoi, it seems that Hanoi is little impacted by this activity. This is probably because rice fields are mostly located far from the city. Thus, owing to the wind and the dispersion of pollutants in the air, the effect of rice straw burning on the air quality of the capital is minute.

It therefore seems that there is **no evidence of the impact of rice straw burning on the air pollution in Hanoi.**

Traffic impact

Let's now spotlight the pollution trends of high traffic density within the city. The most obvious variation in traffic induced pollution are the rush hours. Indeed, twice a day, around 8am and 6pm, a peak of vehicle-related emissions happens. The second less obvious factor of vehicle related pollution are holidays. Interestingly enough, air quality often improves during major holidays (e.g, Tet, the Lunar New Year) when people leave the capital city to return to their hometown. Therefore, during these periods, there is less traffic, and industrial activity slows, resulting in a temporary drop in pollutant concentrations.

On *Figure 15*, we can analyze a typical day of PM10 and PM2.5 emission, and see how it can relate to the aforementioned impact of vehicle traffic.

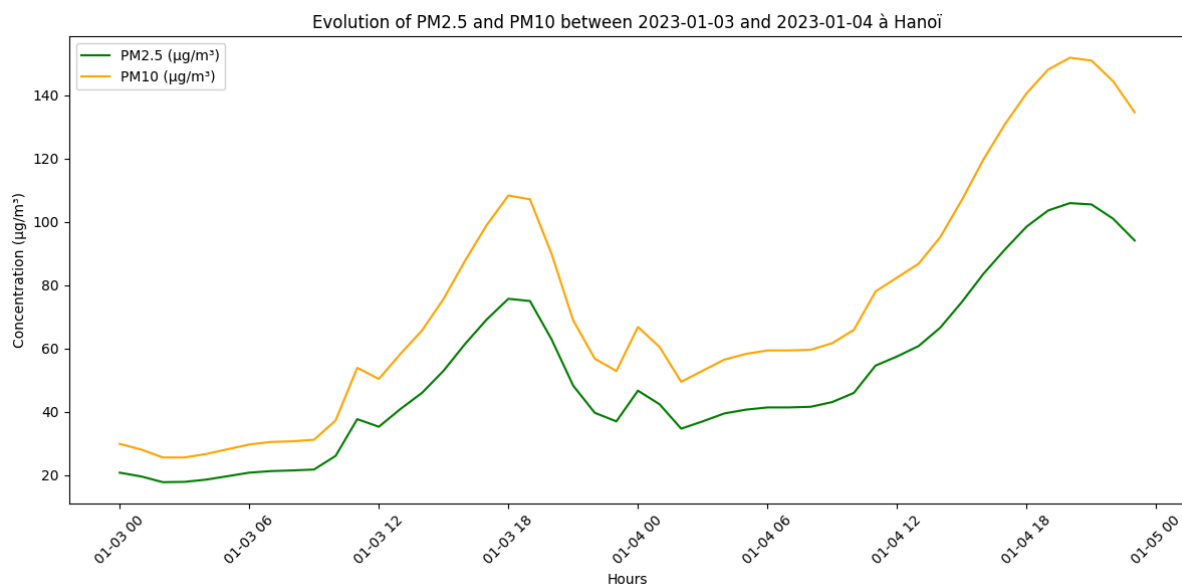


Figure 15 : Daily evolution of PMs' concentration

On a typical day, there is an all time high around 6pm, followed by a decrease in PMs' concentration through the night. However, it is hard to highlight the clear impact of the two rush hours on the PMs concentration, as the air pollution just keeps rising through the day, because pollutants are stagnating in the city. What underscores an increase in PMs emission is the slope of the concentration curve. Indeed we can see that on the two days taken as example, the slope of the curve is steeper around 10am, spotlighting an increase in PMs release. It provides evidence that the morning rush hour happens around this time. Yet, the evening rush hour impact is unclear as the slope of the curve stays rather constant after 11am until 6pm where it reaches its all time high.

Green coverage

Let's now assess the impact of green coverage on air pollution. Intuitively, and even empirically, there should be a negative correlation between green coverage in a city and its air pollution, as vegetation attenuate air PM concentrations via PM2.5 and PM10 deposition on the leaves and alter the dispersion speed of PM2.5/10 in the air.

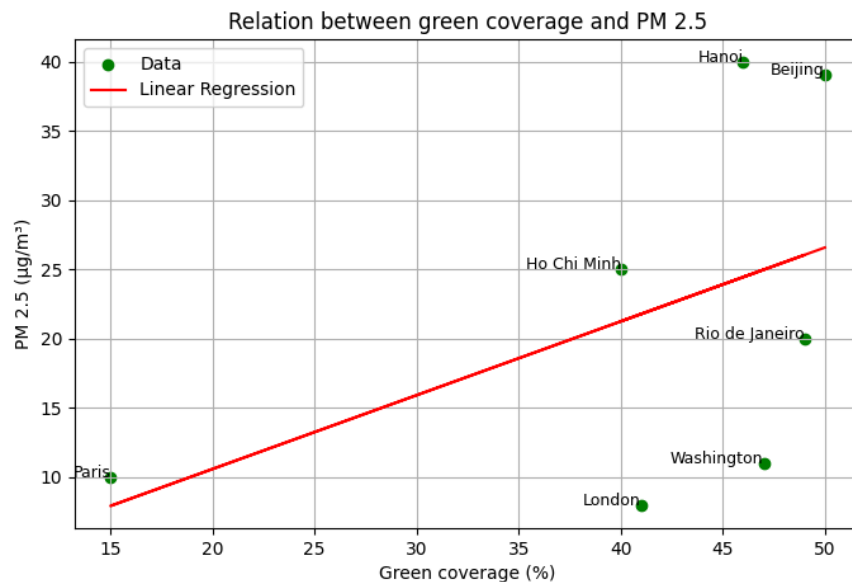


Figure 16 : PM2.5 concentration compare to green coverage

On the above regression with 7 different cities world-wide, there is a positive correlation between green spaces and PM2.5 concentration. Intuitively but also with hindsight, by looking at comprehensive studies on this matter (<https://www.sciencedirect.com/science/article/pii/S1470160X23000237>), we can easily say that this positive correlation is not accurate and shouldn't be taken into account.

In order to have a more accurate reflection of what is happening, we should account for the different other factors in different cities, in order to minimize them, such as population, heavy industries, daily commutation etc ... There is obviously a lack of data on this part, because available green coverage databases are only focusing on European cities, making it hard to do meaningful comparisons with Hanoi.

This leads us to conclude that while green coverage contributes to reducing air pollution in a city, all the other factors play a more significant role. If green coverage were the dominant influence, its effect would outweigh other factors, and a negative correlation would be observed. However high the impact of green coverage might be, it cannot be observed with such a scarcity of green coverage databases in Asia.

Population Impact

Finally, let's now deal with the aggregate impact inhabitants have on air quality in the city.

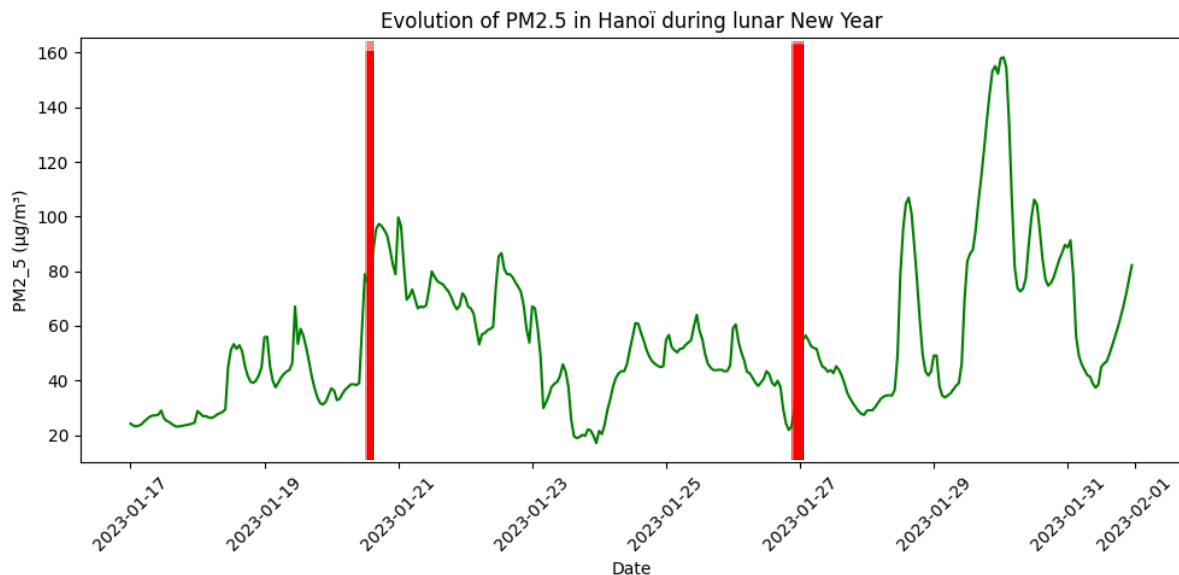


Figure 17 : PM2.5's evolution concentration during January in Hanoi

On the *Figure 17*, we can see that after the Lunar New Year holidays, (which take place from January 21 to 27) the average PM2.5 concentration is notably higher. In contrast, during the holidays, the concentration is lower on average. Lastly, PM2.5 levels increase steadily following the end of the holidays as residents return to Hanoi. This highlights the intrinsic impact that the population's activities can have on PM2.5 concentrations in the city.

3. Conclusion

In this report, we have explored the multifaceted issue of air pollution in Hanoi, analyzing various contributing factors and how they impact air quality across different seasons. By examining data on PM2.5 and PM10 levels, we have highlighted the primary sources of pollution and observed that meteorological elements, such as wind speed, rainfall, play a considerable role in modulating pollution, especially during the monsoon and dry seasons. Our analysis also underscored the complex interplay of these factors, showing that while industrial and vehicular pollution remain constant sources, seasonal and temporal factors can either alleviate or exacerbate pollution levels.

This analysis provides valuable insights that can help Hanoi's residents and policymakers better understand the timing and causes of air quality fluctuations. By focusing on specific periods of heightened pollution, authorities could implement policies or awareness campaigns to mitigate exposure and protect public health. The results of this report can help highlight the need for continued monitoring, stricter pollution controls, and sustainable urban planning practices that address the root causes of air pollution.